

reflectors for concentration and ultra-lightweight but efficient photovoltaic arrays that can be easily maintained and repaired for energy collection and conversion to electricity.

For construction, using lunar materials, for example, could be a convenient option in space. The completed array would orbit above Earth and beam energy back to ground-based receivers. Effective transmission is possible only with efficient wireless systems. Laser beaming and microwave power transmission are typical preferences due to lower weight, cost as well as high efficiency.

For the orbit, choosing geosynchronous orbit (GEO) offers multiple advantages including extensibility and

almost constant power transmission, but the tough part is dealing with radiation exposure. On Earth, energy can be received by rectenna, a rectifying antenna used in wireless systems that transmit power by radio waves.

Thanks to companies like SpaceX and Blue Origin, cost of space launch and hardware are slowly decreasing. Developments in modular manufacturing and robotics are enabling mass production of small modular systems that bring down satellite costs.

In 2018, scientists from California Institute of Technology created a tile weighing just 1.5 kilograms per square-metre, which could be integrated to form large solar arrays for harnessing solar energy.

While attempting it, risks posed to the planet, including living organisms, need to be taken into consideration. Safety protocols such as for transmission power levels of microwaves need to be followed by nations. Since the spectrum that transmits RF waves is already congested, band allocation and geopolitical challenges may arise. Beaming a large amount of power can lead to increased surface temperature. Other space hazards such as solar flares also need to be taken into account.

There is still a long way to go but if appropriate time and resources are available, space-based solar power can offer innumerable benefits for mankind.

—Ayushee Sharma

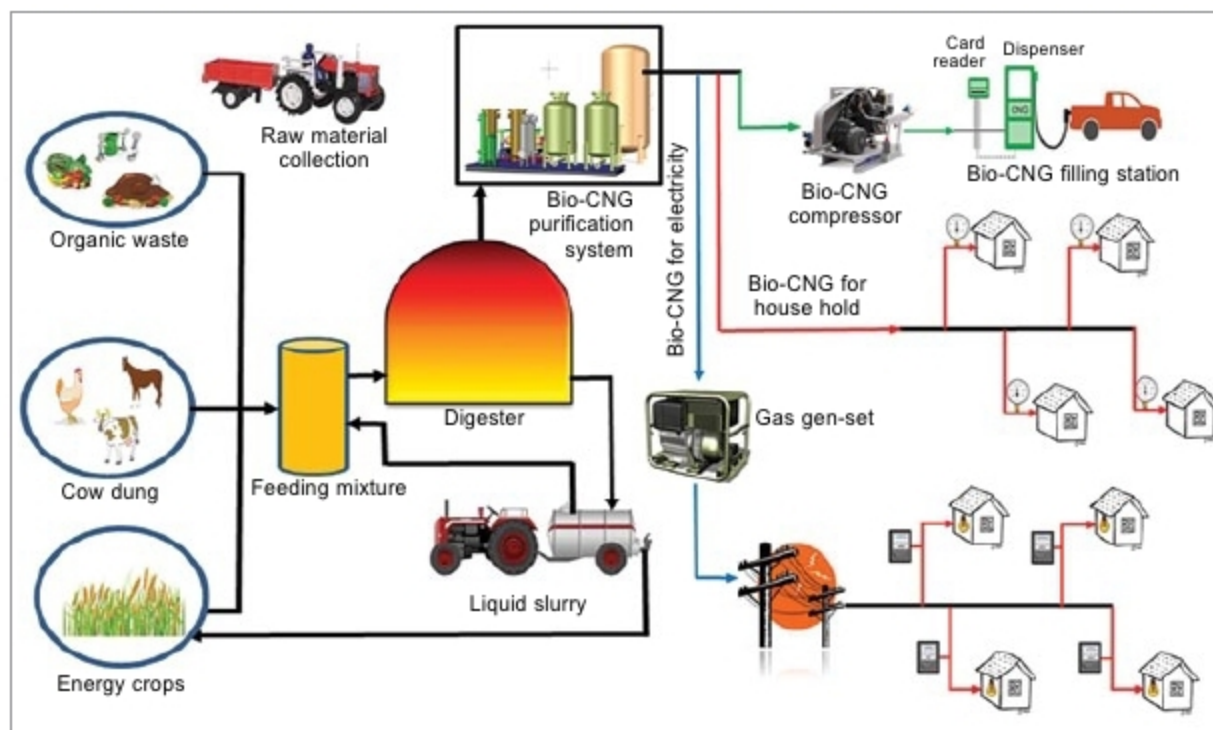
2 BIO-CNG: THE PANACEA FOR GREEN TRANSPORTATION?

Global adoption of bio-CNG depends on several economic, environmental and technical factors

Transport sector is one of the major contributors to greenhouse gases on the planet. As per UN World Meteorological Organization, global temperatures will rise by 3.5 degree Celsius in this century if considerable measures are not adopted. Nations have been focusing on utilising renewable energy sources as alternative fuel sources to reduce the environmental impact. But questions arise. Upscaling of which source is the most effective alternative? Could cow dung or vegetable peels be the ultimate answer to this problem?

A recent study by French Institute of Petroleum (IFPEN) has revealed that the biogas sector can contribute to de-carbonising transportation. The study collated carbon footprint data of the complete lifecycle of compressed natural gas (CNG), biomethane, gasoline, diesel and electric vehicles (EVs), and concluded that biomethane is the best option for maintaining air quality. It recommends a mix of biomethane and CNG till 2030 to power vehicles with a climate impact equivalent to that of an EV, as a mid-term solution.

Agricultural residues, food waste,



Biogas, a simple layout (Credit: www.cbsenergy.com)

organic municipal waste, bio-waste from industries and so on are common sources of biomass. Raw biogas consisting of methane, carbon-dioxide and such impurities as hydrogen-sulphide and water vapour is produced from biomass in the absence of oxygen, and can be converted into different automobile fuels like bio compressed natural gas (bio-CNG) or

compressed biogas (CBG), gasoline, syngas and liquefied biogas.

Among these, bio-CNG offers numerous advantages as a renewable vehicle fuel, such as high calorific value and cost savings over conventional fuels. It is similar to natural gas in terms of composition and properties.

Biogas, when purified to get